PIPING ARRANGEMENT EDIT

ACE Arrangement Edit (E3D)

E3D arrangement (plan) 200-series pipeline — centerline, symbols, line-no stretch, dimensioning, tag-no arrange, with option dialog.

Key Purpose

Why this program

1. **Auto-finish raw E3D arrangement DXF into an issue-ready plan**

   – One Apply runs the full 200-series chain — centerline, symbols, line-no stretch, dimensioning, tag arrange

   → **Effect:** Hours of manual drafting cleanup per plan collapse to a single click, with consistent output every time

2. **Read the drawing and pre-set the options for you**

   – Scans layers/text to detect elevation mode, drawing scale and which tags (line/valve/inst/equip/stru) exist

   → **Effect:** The engineer just picks files — correct tagging is auto-checked, so nothing is missed and nothing is over-drawn

3. **Standardise annotation, dimension and font rules across every sheet**

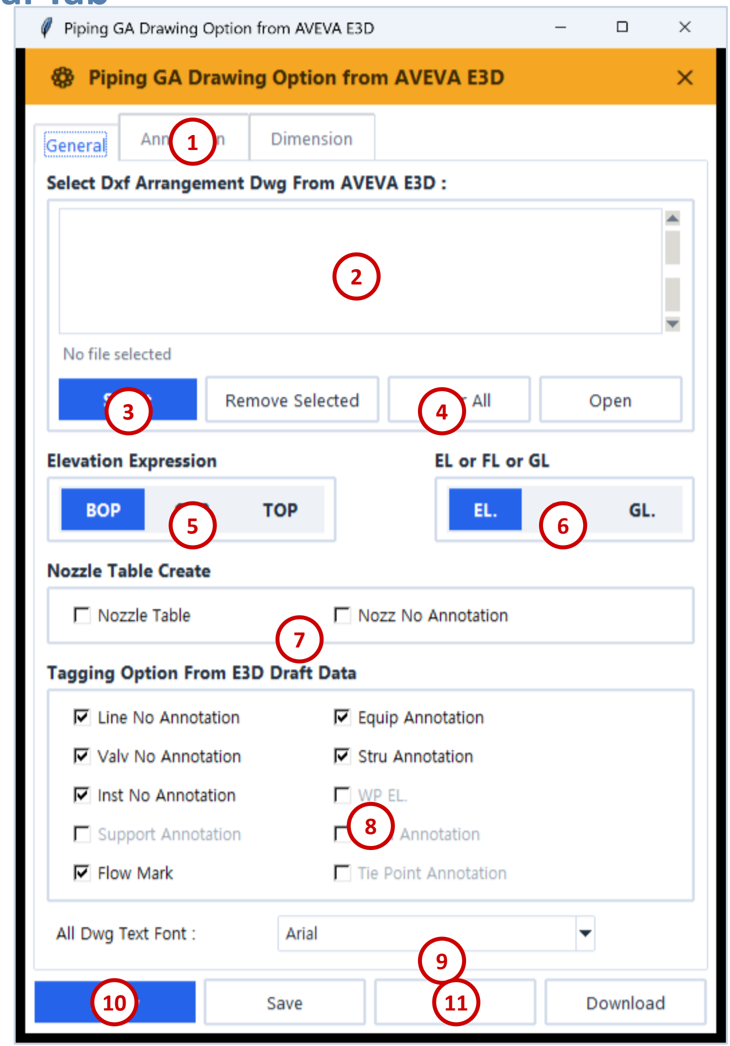
   – Central control of tag symbols, dim colors/gaps, elevation prefix and a single unified drawing font

   → **Effect:** Every plan follows the same house style without hand-editing, cutting review comments and rework

> **Required inputs :** One or more AVEVA E3D-exported arrangement (plan) DXF drawings

Main Window – Item Guide

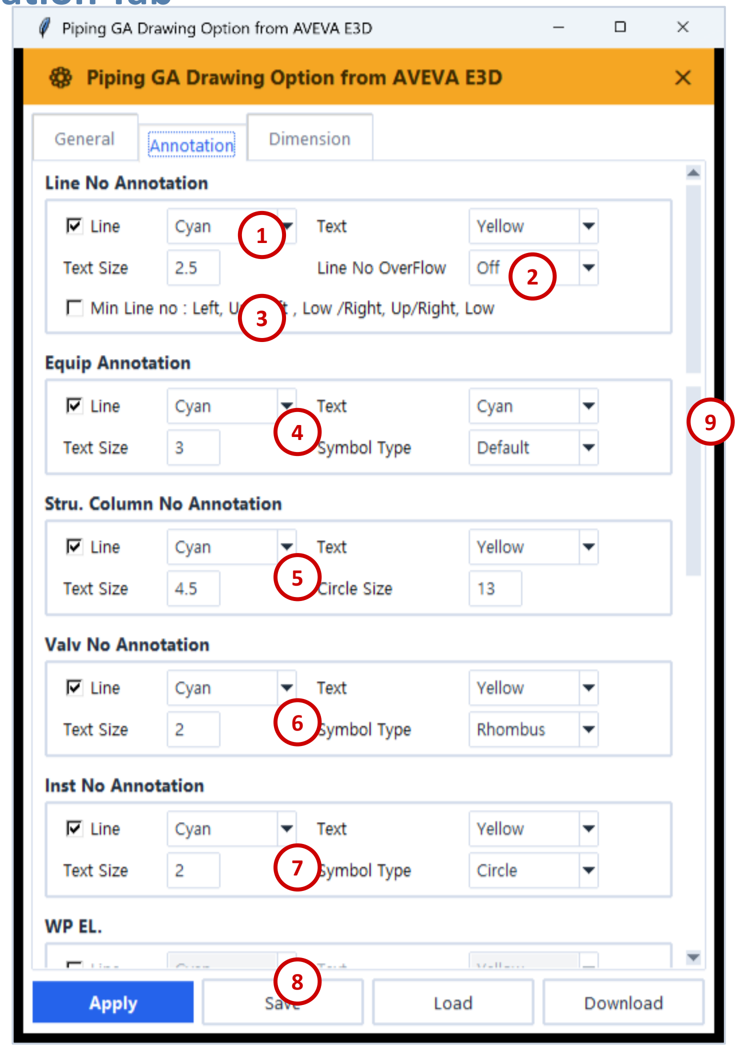
General Tab



1. Tab selector: General / Annotation / Dimension
2. Selected arrangement DXF file list
3. Select DXF drawings
4. Remove Selected / Clear All / Open
5. Elevation Expression: BOP / COP / TOP
6. Elevation prefix: EL / FL / GL
7. Nozzle Table + Nozz No Annotation checkboxes
8. Tagging Option checkboxes (auto-detected)
9. All Dwg Text Font (unify to one font)
10. Apply — run full edit chain
11. Save / Load options, Download results

Main Window – Item Guide

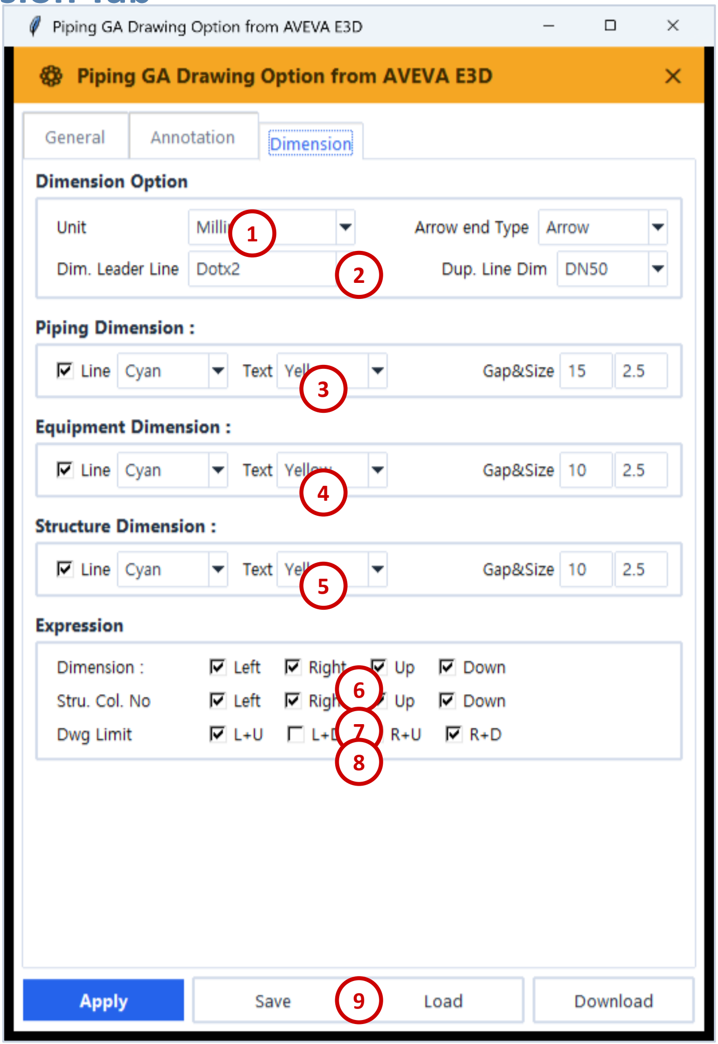
Annotation Tab



1. Line No: line/text color and text size
2. Line No OverFlow toggle
3. Min Line no corner-keep checkbox
4. Equip Annotation: colors, size, symbol type
5. Stru. Column No: colors, size, circle size
6. Valv No Annotation: Rhombus symbol type
7. Inst No Annotation: Circle symbol type
8. WP EL. block (scroll for more blocks)
9. Scrollbar for annotation blocks

Main Window – Item Guide

Dimension Tab



1. Unit + Arrow end Type selectors
2. Dim. Leader Line + Dup. Line Dim selectors
3. Piping Dimension: colors + Gap&Size
4. Equipment Dimension: colors + Gap&Size
5. Structure Dimension: colors + Gap&Size
6. Dimension sides: Left / Right / Up / Down
7. Stru. Col. No sides: Left / Right / Up / Down
8. Dwg Limit corners: L+U / L+D / R+U / R+D
9. Apply / Save / Load / Download buttons

Processing Steps

What 'Apply' runs, in order

1. Assign E3D layers
2. Create centerlines
3. Apply drawing settings
4. Edit symbols
5. Delete duplicate line-no
6. Stretch line-no leaders
7. Create dimensions
8. Edit dimensions
9. Edit tag numbers
10. Arrange tag numbers
11. Unify drawing font
12. Purge unused resources
13. Save as _ALL.DXF

Overview

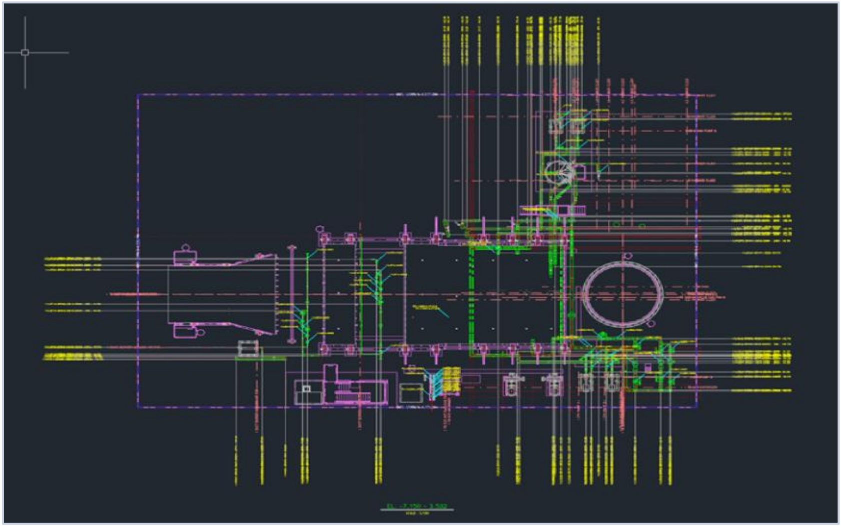
What it does

1. **E3D arrangement (plan) 200-series pipeline — centerline**
2. **Symbols**
3. **Line-no stretch**
4. **Dimensioning**
5. **Tag-no arrange**
6. **With option dialog**

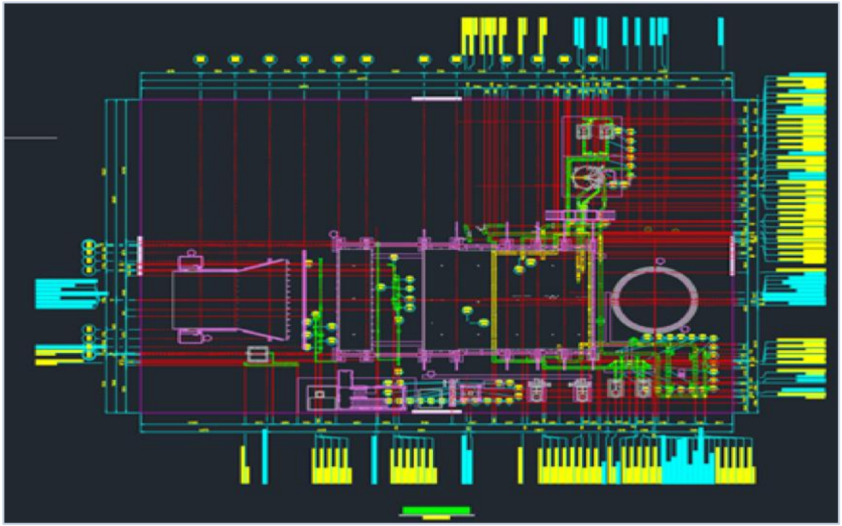
Category: Piping Arrangement Edit
Folder: PIPING_ARRANGEMENT_EDIT/ACE_E3D
Main scripts: ACE_RUNALL.PY
Scripts / Status: 14 scripts · Operational
Tags: E3D · 평면도 · 200번대 · Tkinter

Output Samples

Output Samples (1/3)



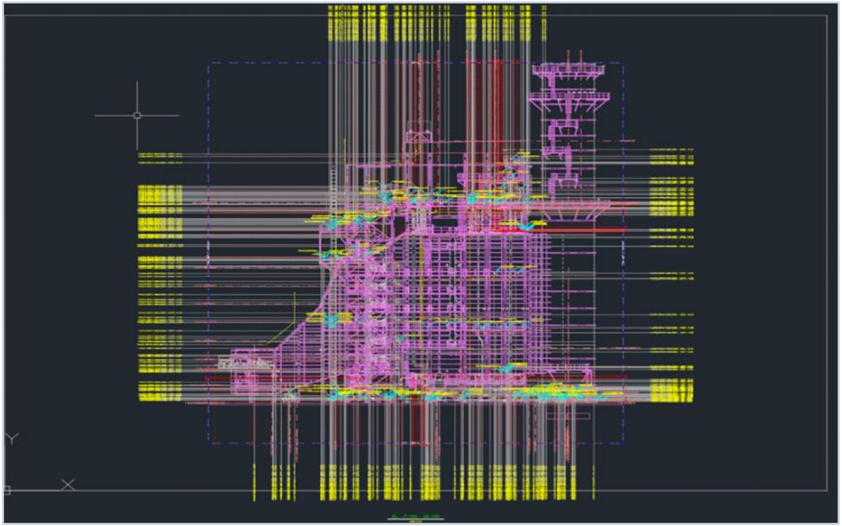
E3D Input File for Piping GA 1



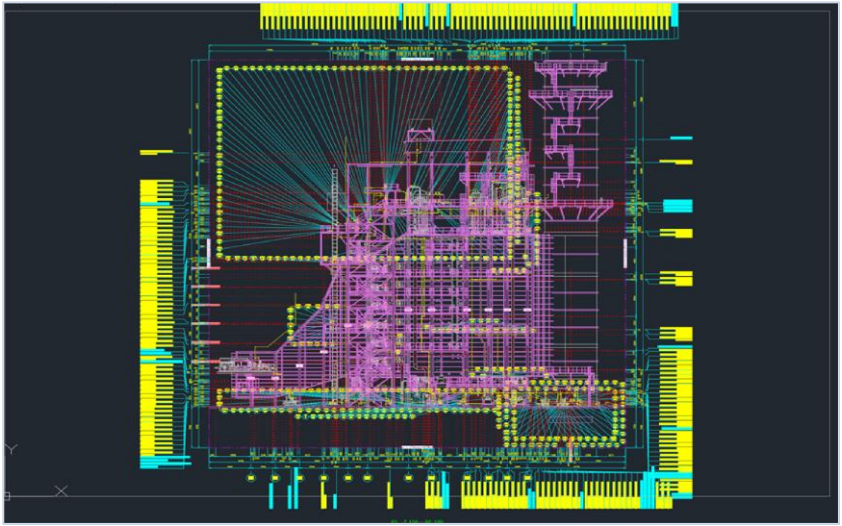
E3D Auto Output file for Piping GA 1

Output Samples

Output Samples (2/3)



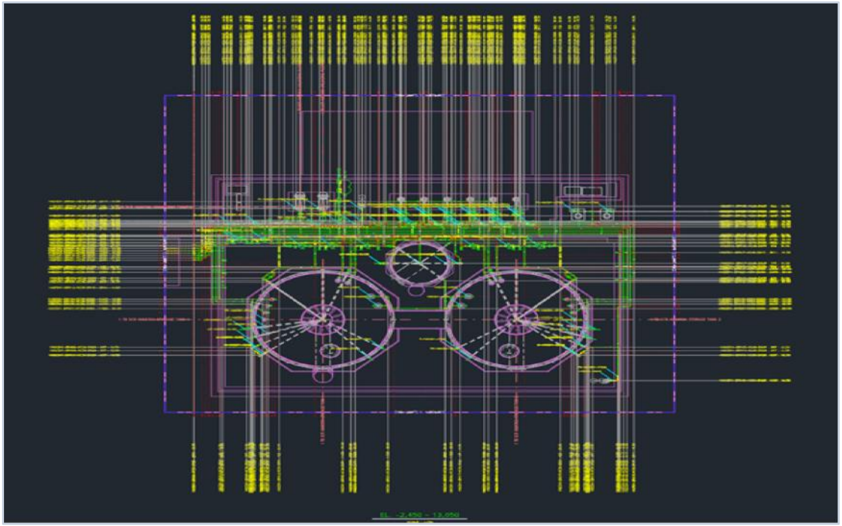
E3D Input File for Piping GA 2



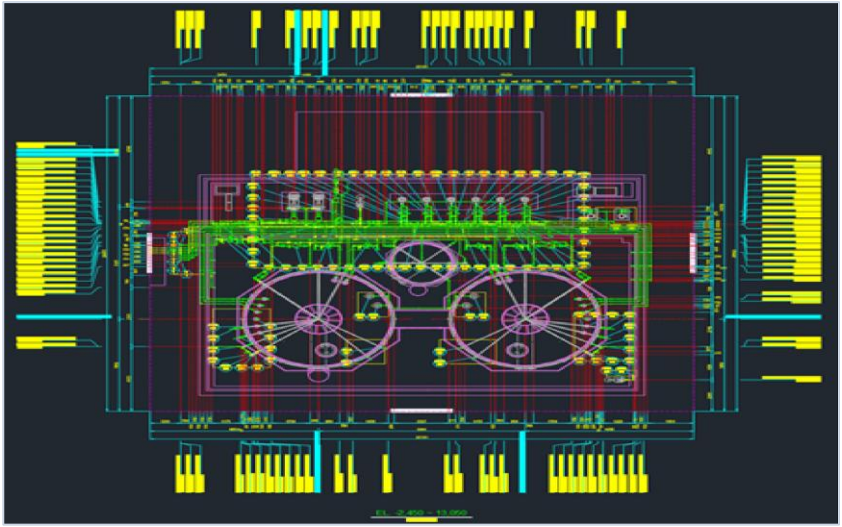
E3D Auto Output file for Piping GA 2

Output Samples

Output Samples (3/3)



E3D Input File for Piping GA 3



E3D Auto Output file for Piping GA 3

Notes & Information

- Part of the in-house Plant Engineering automation suite — category: Piping Arrangement Edit.
- Runs offline as a pure-Python / Tkinter tool; portable across PCs via OneDrive sync (needs Python + packages).
- Launch from the PC launcher (런처.bat) or run the main script: ACE_RUNALL.PY.
- Output samples and the program window above reflect the current build.

Thank You

ACE Arrangement Edit (E3D) · Plant Engineering Total Service System